SHUTAO ZHENG

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WORK EXPERIENCE

Full Stack Engineer | AEC Daily Inc | Newmarket, ON **May 2024 - Present**

· Led the development and deployment of multiple white-label compliance-training portals, using hostname-based routing to deliver client-specific branding and configuration while sharing centralized course data and core application functionality; owned PostgreSQL provisioning, release workflows, and junior developer support.

· Designed and delivered a multilingual localization system using ordered language preferences, graceful fallback logic, and gettext-based template rendering, unifying course-content and interface translations while building pre-release checks to detect untranslated, missing, or malformed strings.

· Remediated two SQL injection vulnerabilities and a privilege-escalation flaw by replacing client-controlled identity inputs and email-domain authorization with server-side session data and database-backed access controls.

· Audited compliance-critical deadline and completion logic, identified reproducible data-integrity issues, and centralized four scattered request-validation points into a consistent application-boundary security control.

Software Engineer | Lifion by ADP | New York, NY **May - Aug. 2022**

· Extended an internal low-code workflow engine used by non-technical staff to configure server-side business logic, adding conditional operators that expanded the range of business rules users could define without writing code.

· Redesigned the application's results interface to display more detailed workflow execution outputs, improving visibility into runtime behavior and helping users validate and troubleshoot configured workflows.

· Developed integration tests within the team's Jenkins CI pipeline, achieving over 90% test coverage for newly introduced functionality and reducing regression risk during subsequent releases.

Quality Assurance Engineer | NetDragon Websoft Holdings | Fuzhou, China **Sept. 2020 - Aug. 2021**

· Built an automated UI testing framework for ActivPanel that simulated user interactions and validated on-screen responses, detecting functional defects before production release and improving confidence in product stability.

· Executed regression and feature testing across three product launches, documented and tracked defects in JIRA, and prepared release-readiness reports to support go/no-go decisions and prioritize critical issues before deployment.

PROJECTS

CELPIP Exam Practice Platform www.deercelpip.com **JavaScript • Python • PostgreSQL • Git**

· Architected a production English exam practice platform with 30+ Flask REST APIs, powering test delivery, user submissions, AI-based scoring, personalized feedback, and speech-to-text transcription.

· Designed PostgreSQL schemas, indexing, and transaction handling to maintain data consistency under concurrent submissions, supporting thousands of learners with sub-second response times.

· Secured the platform end-to-end with JWT authentication, PBKDF2 password hashing, Fernet encryption, and defenses against SQL injection and XSS.

· Reduced page-load time by 60% through CDN delivery, caching, and compression; validated platform stability under 500+ concurrent users through load testing.

Couples Calendar App **Flutter • Dart • Swift/WidgetKit • CloudBase Serverless**

· Built a local-first Flutter application using Drift (SQLite) with incremental Tencent CloudBase synchronization, implementing last-write-wins conflict resolution to reconcile offline edits across calendar, mood, and journal modules.

· Developed CloudBase serverless functions handling authentication, data synchronization, and partner-interaction features such as habit-tracking check-ins and notifications.

· Created native iOS Home Screen Widgets with Swift and WidgetKit and automated iOS builds and releases through Xcode Cloud.

Personal Portfolio Website stzheng.com **React • JavaScript • Figma • HTML/CSS**

· Designed and built a responsive personal portfolio in React and Figma to showcase professional experience and projects.

EDUCATION

University of Waterloo **Waterloo, Ontario**

Bachelor of Mathematics September 2018 - August 2023

Computational Mathematics Major & Combinatorics and Optimization Minor & Computing Minor

SKILLS

Programming Languages: Python, PHP, JavaScript, Dart, Swift, SQL, C++, HTML/CSS

Frameworks and Libraries: React, Flask, Node.js, Flutter, Smarty, WidgetKit

Development Tools: Git, Figma, Docker, Linux, PostgreSQL, Jenkins, Xcode Cloud, Jira, Bugzilla